

PROGRAM 22

AIM

To implement a Family Tree using Prolog and retrieve relationships such as parent, child, sibling, and grandparent.

ALGORITHM

1. Start the program.
2. Define facts for the family members and their parent relationships.
3. Define rules for parent, child, sibling, and grandparent relationships.
4. Load the Prolog program.
5. Enter a query to find a particular relationship.
6. Prolog matches the query with the available facts and rules.
7. Display the result of the query.
8. Stop the program.

PROGRAM

```
male(john).  
male(paul).  
male(mike).  
male(david).  
female(mary).  
female(susan).  
female(lisa).  
female(anna).  
parent(john, paul).  
parent(mary, paul).  
parent(john, susan).  
parent(mary, susan).  
parent(paul, david).  
parent(lisa, david).  
parent(paul, anna).
```

```
parent(lisa, anna).  
father(X,Y) :- male(X), parent(X,Y).  
mother(X,Y) :- female(X), parent(X,Y).  
child(X,Y) :- parent(Y,X).  
sibling(X,Y) :- parent(Z,X), parent(Z,Y), X \= Y.  
grandparent(X,Y) :- parent(X,Z), parent(Z,Y).
```

OUTPUT

```
30 ?- consult(['c:/Users/Hema Neepe/OneDrive/Documents/Family_Tree.pl']).  
true.  
  
31 ?- father(john, paul).  
true.  
  
32 ?- sibling(paul, susan).  
true  
Unknown action: g (h for help)  
Action? ;  
true.  
  
33 ?- andparent(john, david).  
ERROR: Unknown procedure: andparent/2 (DWIM could not correct goal)  
34 ?- grandparent(john, david).  
true .
```

RESULT

The Family Tree was successfully implemented in Prolog, and the required family relationships were obtained by executing suitable queries.